

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

GENERATIVE AI AND APPLICATIONS [CSE-40A3]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Outline the need of Generative AI models in Artificial Intelligence in different domains. (5)
 - A)
 - B) Explain about Autoregressive models and discuss about their limitations and challenges in detail. (5)
- 2) An e-commerce company wants to enhance the shopping experience by generating personalized product images based on customer preferences. The company has a dataset of clothing images, along with metadata such as color, style, and category. Instead of relying only on existing products, the team decides to use a Variational Autoencoder (VAE) to generate new clothing designs that match a user's browsing history and preferences. (5)
 - A) Develop a Python script/pseudocode using TensorFlow/Keras that:
 - Loads and preprocesses a dataset of fashion product images, such as the Fashion-MNIST dataset or the DeepFashion dataset.
 - Builds a Variational Autoencoder (VAE) where the encoder maps images into a latent probability distribution, and the decoder reconstructs new images from this distribution.
 - B) Use the above scenario in Question 2A and answer the following by developing a python script/pseudocode. (5)
 - Train the model using Reconstruction Loss (MSE) and Kullback-Leibler (KL) Divergence to create meaningful latent representations.
 - Evaluate the model by generating new product images and visualizing reconstructed vs. original images.
- 3) An animation studio wants to automate the creation of unique anime character faces to assist designers in developing new character concepts. Instead of manually drawing every face, the team decides to use a Deep Convolutional GAN (DCGAN) to generate high-quality, anime-style character portraits. The goal is to train a model that learns the distribution of anime faces and produces new, visually appealing characters that (5)
 - A)

resemble real anime art.

Develop a Python script/pseudocode using TensorFlow/Keras that:

- Loads and preprocesses a dataset of anime faces, such as the Anime Faces Dataset (e.g., from Getchu or Danbooru) or the CelebA-HQ dataset (if using real human faces for anime-style adaptation).
- Builds a Deep Convolutional GAN (DCGAN) consisting of:
- A Generator that takes random noise as input and generates 64x64 or 128x128 anime character images using transposed convolutions.
- A Discriminator that classifies images as real (from the dataset) or fake (generated).

B) Use the above scenario in Question 3A and answer the following by developing a python script/pseudocode.

- Train the DCGAN using Binary Cross-Entropy loss and the Adam optimizer, ensuring that the generator improves in producing realistic anime faces. (5)
- Evaluate the model by displaying sample-generated characters and comparing them with real ones.

4) A meteorological agency wants to automate the generation of natural language weather reports from structured sensor data. Currently, forecasters manually translate temperature, humidity, wind speed, and other metrics into human-readable reports. To streamline this process, the agency decides to use a Recurrent Neural Network (RNN) model that can learn from historical weather data and corresponding text descriptions to generate real-time, readable forecasts.

A)

Develop a Python script/pseudocode using TensorFlow/Keras that:

- Loads and preprocesses a dataset containing structured weather data (e.g., temperature, wind speed, humidity) and their corresponding text-based weather reports. Suitable datasets include the NOAA Weather Dataset or custom weather text pair corpora. (5)
- Builds an RNN-based sequence-to-sequence model, consisting of:
- An encoder that processes numerical input features (weather parameters).
- A decoder that uses LSTM or GRU layers to generate corresponding textual weather descriptions.
- A Dense layer with SoftMax activation for word prediction.

B) Use the above scenario in Question 4A and answer the following by developing a python script/pseudocode. (5)

- Train the model using Categorical Cross-Entropy loss and the Adam optimizer, ensuring it learns to produce fluent and accurate reports.
- Evaluate the model by generating weather summaries for unseen weather data and comparing them with actual human-written reports.

- 5) Illustrate two emerging trends and opportunities in the field of prompt engineering. (5)
- A)
- B) Explain the key differences between Large Language Models and traditional NLP models. (5)

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